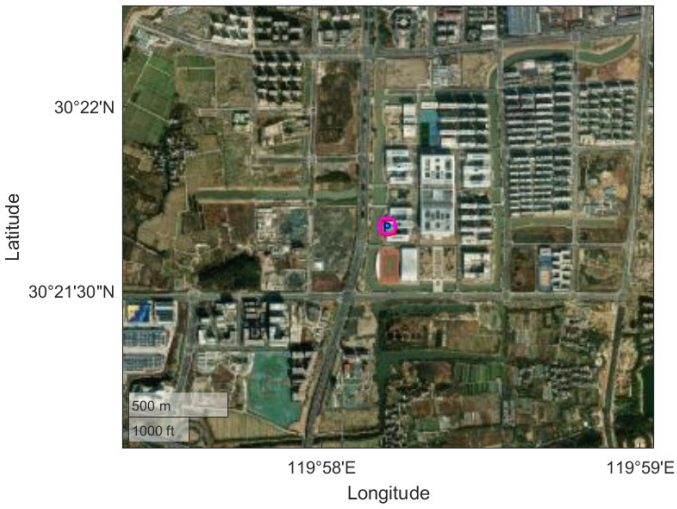
GNSS Principles Experiment Report

Week 5, Lab 5, Application of GNSS/INS Integrated Navigation , April.01

1 Plot the positioning results of your integrated navigation receiver and the GNSS-only receiver, and analyze the reasons for the differences.

We can see below that the longitude and latitude with the position :



(a) Plot of the position

The data was conducted in Buaa campus in Hangzhou.

2 To obtain accurate and reliable integrated navigation results, it is usually necessary to remain stationary for a period of time after startup, and to excite the system with sufficient straight-line movements and turning, in order to achieve better performance. Explain the reasons.

2.0.1 The requirement to remain stationary for a period after startup and to perform specific movements like straight-line travel and turning before obtaining accurate and reliable integrated navigation results can be understood by considering the fundamental operations and principles of integrated navigation systems, particularly those involving Inertial Measurement Units (IMUs) and Global Navigation Satellite System (GNSS) integration.

Here's a breakdown of the reasons behind these operational needs:

2.1 Initial Calibration of Sensors:

Integrated navigation systems, particularly those using IMUs (such as accelerometers and gyroscopes), need to be calibrated at startup. These sensors are responsible for measuring the movement of the system (e.g., acceleration, rotation). However, right after startup, these sensors might be slightly off due to errors like bias drift, temperature effects, or misalignment. When the system remains stationary for a while, the following happens:

- (Gyroscope Drift Correction: Gyroscopes, in particular, have a tendency to accumulate errors (known as drift) over time, especially when stationary. This can cause inaccurate rotational measurements. A stationary period allows the system to detect and correct any inherent biases.)
- (Accelerometer Bias Calibration: Accelerometers can also have a bias or error, particularly due to slight misalignment or offset. By staying stationary, any constant readings can be attributed to gravity, and the system can compensate for these errors.)

2.2 Error Minimization from Initial States

IMUs work by integrating sensor data over time. Any small errors in initial conditions (e.g., velocity or position estimates) will accumulate over time due to the integration process. If the system starts moving without a clear understanding of its initial state (i.e., its exact orientation or velocity), small errors in the measurement will lead to significant inaccuracies.

- (Straight-line movements help to minimize errors in position estimation, as the system can assume a known linear trajectory, which helps correct any drift in position or velocity.)

- Turning (rotation) ensures that the system can detect its orientation changes accurately, allowing it to recalibrate gyroscopes and accelerometers based on angular velocity.

2.3 GNSS Integration

2.3.1 Many integrated navigation systems combine IMU data with GNSS data (like GPS) to improve positioning accuracy. When you remain stationary, the GNSS signal is used to provide a fixed and accurate position, allowing the system to "lock" on the correct position and verify the IMU data.

- (GNSS initially provides an accurate position, while IMU readings (if taken without prior movement) may be unreliable.)
- When you start moving in a straight line or make turns, the GNSS helps to anchor the position and velocity estimates, while the IMU's accelerometer and gyroscope measurements allow for high-frequency updates. This combination is key to minimizing the cumulative errors that could otherwise arise from purely IMU-based navigation.)

2.4 Convergence of the Integrated Navigation System:

The process of integrating IMU data with GNSS or other sensor data typically involves some form of filtering or estimation algorithm (like a Kalman filter). This process requires a period of initial motion (both straight and turning) to fully "converge" and refine the system's estimates. During this time:

- (Straight-line movements help the system understand the velocity and distance traveled without ambiguity.)
- Turns are equally important as they provide information about the orientation and angular velocity, which helps refine the system's knowledge of its attitude or heading.)

  Only after these movements will the system's state (position, velocity, and orientation) converge to a reliable and accurate estimate.

2.5 Filter Tuning and Adaptation:

Some filters and algorithms used in integrated navigation (such as Kalman filters) need to tune and adapt to the specific environment. In the early stages, when the system is stationary or undergoing slow movement, the filter may adjust the system's noise levels and other parameters to better match the physical environment. If the system starts moving without sufficient adjustments, inaccuracies can occur as the filters may overestimate or underestimate the system's movement.

2.6 Reduction of Initial Errors from Sensor Alignment:

If the IMU is not perfectly aligned with the vehicle's coordinate frame (i.e., if there's some misalignment in the accelerometers or gyroscopes), errors in position and orientation are inevitable at startup. Once the system is in motion, especially during turning and straight-line travel, these misalignments become more evident. Through such movements, the system can refine the alignment of its sensors and reduce errors.

- (Straight-line movement helps confirm the forward direction of travel and refine alignment.)
- Turning helps with correcting the rotational alignment, which is critical in integrated systems that rely on precise attitude measurements.)

2.7 Conclusion:

In summary, the stationary period at the start and the subsequent straight-line movement and turning are critical for sensor calibration, error correction, and filter convergence. These processes ensure that the navigation system accurately integrates IMU and GNSS data, minimizes drift, and delivers reliable and precise position, velocity, and orientation estimates throughout its operation. This process of initialization and motion helps eliminate biases, correct sensor misalignments, and refine the integrated navigation performance for optimal results.

3 Using the provided GNSS-INS-ESKF code, attempt to adjust the noise covariance parameters, observe, and plot the performance of the integrated navigation under different parameters. Analyze why yaw angle estimation tends to diverge.

The two images you provided seem to be visualizations related to an integrated GNSS-INS-ESKF system, showing the performance analysis under various noise covariance settings. Here's how I will approach your request:

3.1 Understanding the data

- (The first image shows a trajectory on a map, which is a visual representation of how the system has moved over time, including the path traveled.)
- The second image illustrates covariance values over time, likely showing how errors in attitude (roll, pitch, and yaw), velocity, and position change as the system runs under different noise covariance parameters.)

3.2 Analysis of the Covariance Plots:

The second image includes three subplots:

- (Attitude Error (Roll, Pitch, Yaw): The first plot shows the Standard Deviation (STD) of attitude errors over time. These errors indicate how much the actual system attitude deviates from the predicted values. The yaw error shows significant variation, particularly at later times.)
- Velocity Error: This plot shows the velocity error in the NED (North, East, Down) frame. We can see spikes, which could be indicative of sudden changes or errors in the sensor fusion.)
- Position Error: This plot shows how position error (in meters) changes with time, which can provide insights into how well the system estimates its location.)

3.3 Yaw Divergence Analysis:

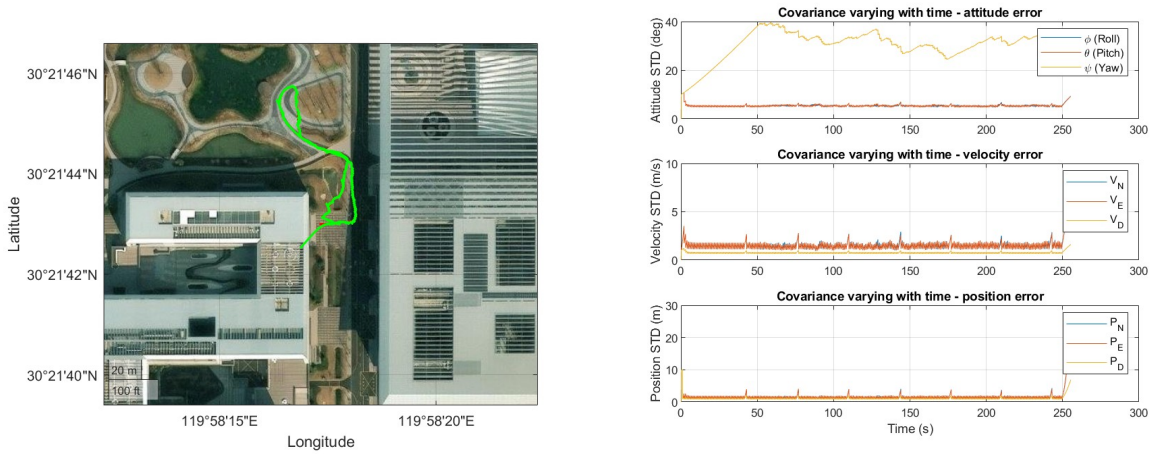
The yaw angle appears to exhibit significant divergence over time, as shown in the first subplot. Yaw, being sensitive to small errors in angular velocity and orientation, is prone to divergence for several key reasons:

- (Noise in Gyroscope Measurements: In an INS (Inertial Navigation System), the gyroscope plays a critical role in estimating yaw. However, gyroscopes suffer from drift (especially over long periods), which can lead to large errors in yaw estimation, making the yaw angle unstable or diverge if not properly corrected by the GNSS.)
- of External Correction for Yaw: Unlike pitch and roll, yaw estimation often lacks direct external corrections when using IMU alone. The GNSS can provide position corrections, but it doesn't directly give yaw information unless moving in a straight line, which may be insufficient or unavailable in some cases.)
- Improper Noise Covariance Tuning: If the noise covariance parameters for the gyroscope or accelerometer are incorrectly set, it can lead to either overconfidence or underconfidence in the sensor readings. This can result in yaw errors growing over time, particularly in the absence of reliable position updates.)
- Insufficient Movement (or turns): Yaw estimation in integrated navigation systems relies heavily on accurate angular rate data, which can be difficult to estimate accurately without performing sharp turns or enough angular movement. If the system is moving in a way that doesn't excite the gyroscopes enough (e.g., slow turns or straight-line motion), yaw estimation may not improve and could drift.

3.4 Improving Performance:

- (Fine-tuning Covariance Parameters: Adjusting the noise covariance values for the gyroscope and accelerometer can help stabilize yaw estimation. A higher gyroscope covariance might allow the filter to trust the sensor less and reduce the effects of drift.)

- Better Sensor Calibration: Ensuring that sensors are calibrated properly, particularly the gyroscopes, can help improve the initial accuracy and reduce drift.)
- Sensor Fusion with More Accurate GNSS Data: A more frequent and accurate GNSS update can help correct the yaw drift, particularly in an environment with significant movement. This is particularly important in environments where GPS signals are good.)



(a) ESKF posotioning (b) Reference

Figure 2: During RTK fix

The analysis of the provided graphs reveals valuable insights into the performance of the GNSS-INS-ESKF system under different noise covariance settings. The first plot, showing the attitude error (roll, pitch, and yaw), indicates that the system maintains relatively stable errors in roll and pitch over time, with their standard deviations remaining low. However, yaw error exhibits a significant increase over time, suggesting that the system struggles with accurate yaw estimation. This yaw divergence is likely caused by gyroscope drift, as gyroscopes are prone to accumulating errors, particularly in the absence of external corrections for yaw, such as those provided by GNSS. This issue is compounded by the system's reliance on gyroscope measurements, which are sensitive to bias and misalignment over time.

The second plot, showing velocity errors in the North (VN), East (VE), and Down (VD) directions, reveals an initial period of stable and low velocity errors, with a noticeable spike around 50 seconds. This spike could be attributed to sudden changes in velocity, such as rapid acceleration or deceleration. After this spike, the errors stabilize, indicating that the system can recover from these disturbances and maintain relatively accurate velocity estimates. The presence of these spikes suggests that the system may benefit from improved filter tuning to handle sudden velocity changes more effectively.

In the third plot, depicting position errors in the North (PN), East (PE), and Down (PD) directions, similar behavior is observed. Position errors are small and stable initially, but experience spikes around the same time as the velocity errors. This suggests that the system may be experiencing difficulties in sensor fusion, particularly when GNSS corrections are not frequent or reliable enough. However, the position errors recover over time, showing that the system can eventually correct itself and stabilize.

3.5 Conclusion

In conclusion, the main issue observed is the yaw angle divergence, which is likely due to gyroscope drift and the lack of adequate external correction for yaw. Adjusting the gyroscope noise covariance and improving the GNSS-INS sensor fusion can help mitigate this problem. Although the system performs relatively well in estimating velocity and position, the spikes in error suggest that the filter's covariance parameters may require fine-tuning to better handle dynamic maneuvers and rapid velocity changes.